1. For a function of two variables z = f(x, y), the partial derivative dz/dx is computed by treating y as:

(A) A variable
(B) A constant
(C) Zero
(D) A function of x

2. Clairaut's (Schwarz's) Theorem states that if f(x, y) and its mixed partials are continuous, then:

(A) f_xx = f_yy
(B) f_xy = f_yx
(C) f_xy = -f_yx
(D) f_x * f_y = 1

3. A function f(x, y) is called homogeneous of degree n if for any scalar t > 0:

(A) f(tx, ty) = t * f(x, y)
(B) f(tx, ty) = t^n * f(x, y)
(C) f(tx, ty) = n * t * f(x, y)
(D) f(t+x, t+y) = t^n

4. Euler's Theorem on Homogeneous Functions states that if u = f(x, y) is homogeneous of degree n, then x*(du/dx) + y*(du/dy) equals:

(A) n
(B) n * u
(C) n * (n - 1) * u
(D) u / n

5. For a homogeneous function u(x, y) of degree n, x^2*(d^2u/dx^2) + 2xy*(d^2u/dxdy) + y^2*(d^2u/dy^2) equals:

(A) n * u
(B) n^2 * u
(C) n * (n - 1) * u
(D) (n - 1) * u

6. The total differential dz of a differentiable function z = f(x, y) is given by:

(A) (dz/dx) + (dz/dy)
(B) (dz/dx)*dx + (dz/dy)*dy
(C) (dz/dx)*dy + (dz/dy)*dx
(D) dx * dy

7. If z = f(x, y) where x = phi(t) and y = psi(t), the total derivative dz/dt is:

(A) (dz/dx)*(dx/dt) + (dz/dy)*(dy/dt)
(B) (dz/dx) + (dz/dy)
(C) (dx/dt) * (dy/dt)
(D) dz / (dx * dy)

8. For an implicit function f(x, y) = c, the derivative dy/dx is given by:

(A) fx / fy
(B) - fx / fy (provided fy != 0)
(C) - fy / fx
(D) fx * fy

9. The Jacobian of u(x, y) and v(x, y) with respect to x and y is denoted J = d(u, v)/d(x, y) and defined as:

(A) ux * vx - uy * vy
(B) det([ux, uy; vx, vy]) = ux * vy - uy * vx
(C) ux + vy
(D) ux * uy * vx * vy

10. If J = d(u, v)/d(x, y) and J' = d(x, y)/d(u, v), the reciprocal property states that:

(A) J + J' = 0
(B) J * J' = 1
(C) J = J'
(D) J * J' = -1

11. Two functions u(x, y) and v(x, y) are functionally dependent if and only if their Jacobian d(u, v)/d(x, y) is:

(A) Positive
(B) Zero
(C) 1
(D) Constant non-zero

12. The Jacobian of transformation from Cartesian (x, y) to polar coordinates (r, theta) where x = r*cos theta, y = r*sin theta is:

(A) r^2
(B) r
(C) 1 / r
(D) r * sin theta

13. The Jacobian of transformation from Cartesian (x, y, z) to spherical polar coordinates (r, theta, phi) is:

(A) r^2
(B) r^2 * sin theta
(C) r * sin^2 theta
(D) r^3 * cos phi

14. The Jacobian of transformation from Cartesian (x, y, z) to cylindrical coordinates (r, theta, z) is:

(A) r
(B) r^2
(C) 1 / r
(D) r * z

15. A point (a, b) is a stationary (critical) point of f(x, y) if:

(A) fx(a, b) = 0 and fy(a, b) = 0
(B) fxx(a, b) = 0
(C) f(a, b) = 0
(D) fxy(a, b) = 0

16. In the second derivative test for f(x, y), let r = fxx, s = fxy, t = fyy. The discriminant Delta is:

(A) r + t - s
(B) r * t - s^2
(C) r * t + s^2
(D) r^2 + s^2 + t^2

17. At a stationary point (a, b), if r*t - s^2 > 0 and r < 0, then f(x, y) has a:

(A) Local Minimum
(B) Local Maximum
(C) Saddle Point
(D) Point of Inflection

18. At a stationary point (a, b), if r*t - s^2 > 0 and r > 0, then f(x, y) has a:

(A) Local Maximum
(B) Local Minimum
(C) Saddle Point
(D) Cusp

19. At a stationary point (a, b), if r*t - s^2 < 0, then the point (a, b) is a:

(A) Local Maximum
(B) Local Minimum
(C) Saddle Point
(D) Extremum

20. At a stationary point (a, b), if r*t - s^2 = 0, the test is:

(A) Definitive for maximum
(B) Definitive for minimum
(C) Inconclusive (further investigation needed)
(D) Saddle point

21. A saddle point of a surface z = f(x, y) is a point where the surface curves:

(A) Upward in all directions
(B) Downward in all directions
(C) Upward in one direction and downward in another
(D) Has zero slope everywhere

22. Lagrange's Method of Undetermined Multipliers is used to find extrema of f(x, y, z) subject to a:

(A) Differential constraint
(B) Functional constraint phi(x, y, z) = 0
(C) Inequality constraint only
(D) Time delay

23. In Lagrange's multiplier method with constraint phi = 0, the auxiliary function F is defined as:

(A) f + lambda * phi
(B) f - lambda / phi
(C) f * phi + lambda
(D) lambda * f + phi

24. The gradient vector grad(f) = [fx, fy, fz]^T at any point on a surface f(x, y, z) = c is:

(A) Tangent to the surface
(B) Normal (perpendicular) to the level surface
(C) Parallel to the z-axis
(D) Zero vector

25. The directional derivative of f(x, y, z) in the direction of a unit vector u is:

(A) grad(f) * u
(B) grad(f) . u (dot product)
(C) grad(f) x u
(D) |grad(f)| / |u|

26. The maximum rate of increase of a scalar function f(x, y, z) occurs in the direction of:

(A) -grad(f)
(B) grad(f)
(C) Tangent plane
(D) z-axis

27. The magnitude of the maximum directional derivative of f at any point is:

(A) 1
(B) |grad(f)|
(C) grad(f) . grad(f)
(D) 0

28. If a measured variable x has small error delta-x, the relative error in x is:

(A) delta-x
(B) delta-x / x
(C) (delta-x / x) * 100
(D) x * delta-x

29. The percentage error in quantity u = x^a * y^b is given by:

(A) a * (dx/x) + b * (dy/y)
(B) [a * (dx/x) + b * (dy/y)] * 100
(C) a * b * (dx/x)
(D) a * dx + b * dy

30. Taylor's theorem for f(x, y) about (a, b) represents f(x, y) as an expansion in powers of:

(A) (x + a) and (y + b)
(B) (x - a) and (y - b)
(C) x * y
(D) x / y

31. The first order terms in the Taylor series of f(x, y) about (a, b) are:

(A) (x - a)*fx(a,b) + (y - b)*fy(a,b)
(B) fx(a,b) + fy(a,b)
(C) (x - a)^2 * fxx
(D) x * fx + y * fy

32. What is the degree of homogeneity of the function f(x, y) = (x^3 + y^3) / (x + y)?

(A) 1
(B) 2
(C) 3
(D) 0

33. What is the degree of homogeneity of f(x, y) = sqrt(x) / (x^2 + y^2)?

(A) 1/2
(B) -3/2
(C) -2
(D) 3/2

34. What is the degree of homogeneity of f(x, y) = sin(x / y)?

(A) 1
(B) 0
(C) -1
(D) pi

35. In thermodynamic bioprocess systems, the Gibbs free energy G(T, P) has partial derivative (dG/dT)_P equal to:

(A) Volume V
(B) -Entropy (-S)
(C) Enthalpy H
(D) Temperature T

36. The partial derivative (dG/dP)_T of Gibbs free energy with respect to pressure at constant temperature is:

(A) Volume V
(B) -Entropy
(C) Internal Energy
(D) -Volume (-V)

37. In bioreactor optimization, the volumetric biomass productivity Q_x = D * X depends on dilution rate D and biomass X. The maximum occurs where:

(A) dQ_x / dD = 0
(B) dD / dX = 0
(C) Q_x = 0
(D) D = 0

38. If z = x^y, then dz/dx is:

(A) y * x^(y - 1)
(B) x^y * ln x
(C) y * x^y
(D) x^(y - 1)

39. If z = x^y, then dz/dy is:

(A) y * x^(y - 1)
(B) x^y * ln x
(C) x * y^(x - 1)
(D) ln(x * y)

40. The limit lim_{(x, y) -> (0, 0)} (x * y) / (x^2 + y^2) along the path y = m*x evaluates to:

(A) 0
(B) m / (1 + m^2) (depends on m, hence does not exist)
(C) 1
(D) Infinity

41. Evaluate lim_{(x, y) -> (0, 0)} (x^2 - y^2) / (x^2 + y^2):

(A) 0
(B) 1
(C) Does not exist (path dependent)
(D) -1

42. Evaluate lim_{(x, y) -> (0, 0)} (x^3 + y^3) / (x^2 + y^2) using polar coordinates (x = r*cos theta, y = r*sin theta):

(A) 1
(B) 0
(C) Does not exist
(D) pi / 4

43. If u = arcsin((x + y) / (sqrt(x) + sqrt(y))), what is x*(du/dx) + y*(du/dy)?
- **(A)** (1/2) * tan u
- **(B)** (1/2) * sin u
- **(C)** tan u
- **(D)** (1/2) * cos u

44. If u = ln((x^4 + y^4) / (x + y)), what is x*(du/dx) + y*(du/dy)?
- **(A)** 3
- **(B)** 3 * u
- **(C)** 4
- **(D)** u / 3

45. If u = (x^2 * y^2) / (x + y), find x*(du/dx) + y*(du/dy):
- **(A)** 3 * u
- **(B)** 4 * u
- **(C)** 2 * u
- **(D)** u

46. If z = f(x + a*y) + g(x - a*y), then d^2z / dy^2 equals:
- **(A)** a * (d^2z / dx^2)
- **(B)** a^2 * (d^2z / dx^2)
- **(C)** -a^2 * (d^2z / dx^2)
- **(D)** (1/a^2) * (d^2z / dx^2)

47. If u = x^2 - y^2 and v = 2xy, find the Jacobian J = d(u, v)/d(x, y):
- **(A)** 2 * (x^2 + y^2)
- **(B)** 4 * (x^2 + y^2)
- **(C)** x^2 + y^2
- **(D)** 4 * (x^2 - y^2)

48. If x = r * cos theta and y = r * sin theta, find the Jacobian J' = d(r, theta)/d(x, y):
- **(A)** r
- **(B)** 1 / r
- **(C)** r^2
- **(D)** 1 / r^2

49. If u = x + y and v = x * y, are u and v functionally dependent?
- **(A)** Yes, always
- **(B)** No, because Jacobian is x - y != 0
- **(C)** Yes, if x = y
- **(D)** Indeterminate

50. If u = (x - y) / (x + y) and v = x * y / (x + y)^2, are u and v functionally dependent?
- **(A)** Yes, because Jacobian d(u, v)/d(x, y) = 0
- **(B)** No, they are independent
- **(C)** Only when x = 0
- **(D)** Only when y = 0

51. Find the stationary points of f(x, y) = x^3 + y^3 - 3axy:
- **(A)** (0, 0) and (a, a)
- **(B)** (a, 0) and (0, a)
- **(C)** (a, -a)
- **(D)** (0, 0) only

52. For f(x, y) = x^3 + y^3 - 3axy (a > 0), the nature of stationary point (0, 0) is:
- **(A)** Local Minimum
- **(B)** Local Maximum
- **(C)** Saddle Point (r*t - s^2 = -9a^2 < 0)
- **(D)** Isolated minimum

53. For f(x, y) = x^3 + y^3 - 3axy (a > 0), the nature of stationary point (a, a) is:
- **(A)** Local Maximum
- **(B)** Local Minimum (r*t - s^2 = 27a^2 > 0, r = 6a > 0)
- **(C)** Saddle Point
- **(D)** Inconclusive

54. Find the minimum value of f(x, y) = x^2 + y^2 + 6x + 12:
- **(A)** 3
- **(B)** 12
- **(C)** 0
- **(D)** -3

55. Find the maximum of f(x, y, z) = x * y * z subject to constraint x + y + z = 3a (x, y, z > 0):
- **(A)** a^3 (at x = y = z = a)
- **(B)** 3 * a^3
- **(C)** a^3 / 3
- **(D)** 27 * a^3

56. Find the minimum distance from the origin (0, 0, 0) to the plane 2x + 3y + z = 14:
- **(A)** sqrt(14)
- **(B)** 14
- **(C)** 2
- **(D)** sqrt(7)

57. If the sides of a rectangular box are measured with 1% error each, the percentage error in the calculated volume V = x*y*z is approximately:

(A) 1%
(B) 2%
(C) 3%
(D) 0.5%

58. If the period of a pendulum is T = 2*pi*sqrt(L / g), a 2% increase in length L causes the period T to:

(A) Increase by 2%
(B) Increase by 1%
(C) Decrease by 1%
(D) Remain unchanged

59. If u = f(x - y, y - z, z - x), then du/dx + du/dy + du/dz equals:

(A) u
(B) 0
(C) 3 * u
(D) x + y + z

60. If x = u*(1 - v) and y = u*v, what is the Jacobian d(x, y)/d(u, v)?

(A) u
(B) -u
(C) v
(D) u * v

61. If u = x + y + z, v = xy + yz + zx, w = x^2 + y^2 + z^2, what is the Jacobian d(u, v, w)/d(x, y, z)?

(A) 1
(B) 0 (since w = u^2 - 2v, functionally dependent)
(C) 2
(D) x * y * z

62. If z = e^(x*y), then d^2z / dxdy equals:

(A) e^(xy)
(B) e^(xy) * (1 + xy)
(C) e^(xy) * xy
(D) x * y * e^(xy)

63. The stationary points of f(x, y) = x^2 + y^2 - 2x - 4y + 5 are:

(A) (1, 2)
(B) (2, 1)
(C) (0, 0)
(D) (-1, -2)

64. At the stationary point (1, 2) of f(x, y) = x^2 + y^2 - 2x - 4y + 5, f attains a:

(A) Local Minimum of 0
(B) Local Maximum of 5
(C) Saddle Point
(D) Local Minimum of 5

65. Find dy/dx if x^3 + y^3 - 3axy = 0:

(A) (ay - x^2) / (y^2 - ax)
(B) (x^2 - ay) / (y^2 - ax)
(C) (y^2 - ax) / (x^2 - ay)
(D) (ax - y^2) / (x^2 - ay)

66. The directional derivative of f(x, y) = x^2 * y at point (1, 2) in the direction of vector v = 3*i + 4*j is:

(A) 20 / 5 = 4
(B) 13 / 5 = 2.6
(C) 11 / 5 = 2.2
(D) 15 / 5 = 3

67. The unit normal vector to the surface x^2 + y^2 + z^2 = 3 at point (1, 1, 1) is:

(A) [1, 1, 1]^T / sqrt(3)
(B) [1, 1, 1]^T
(C) [2, 2, 2]^T
(D) [1, -1, 0]^T

68. The equation of the tangent plane to the sphere x^2 + y^2 + z^2 = 14 at point (1, 2, 3) is:

(A) x + 2y + 3z = 14
(B) x + y + z = 6
(C) 2x + 4y + 6z = 14
(D) x - 2y + 3z = 0

69. The equation of the normal line to the sphere x^2 + y^2 + z^2 = 14 at point (1, 2, 3) is:

(A) (x - 1)/1 = (y - 2)/2 = (z - 3)/3
(B) x + 2y + 3z = 0
(C) (x - 1)/2 = (y - 2)/3
(D) x = 1, y = 2

70. In a continuous stirred-tank bioreactor (CSTR), biomass productivity is P(D) = D * X0 * (1 - D / mu_max). The optimal dilution rate D_opt that maximizes productivity is:

(A) mu_max / 2
(B) mu_max
(C) mu_max / 4
(D) 2 * mu_max

71. The maximum biomass productivity P_max for P(D) = D * X0 * (1 - D / mu_max) is:

(A) mu_max * X0 / 4
(B) mu_max * X0 / 2
(C) mu_max * X0
(D) X0 / 2

72. If u = x^2 / a^2 + y^2 / b^2, then x*(du/dx) + y*(du/dy) equals:

(A) u
(B) 2 * u
(C) 4 * u
(D) 0

73. If u = arctan(y / x), then x*(du/dx) + y*(du/dy) is:

(A) 0 (degree 0)
(B) 1
(C) u
(D) tan u

74. If u = arctan(y / x), then d^2u/dx^2 + d^2u/dy^2 (Laplacian of u) equals:

(A) 0 (harmonic function)
(B) 1
(C) 2
(D) -1 / (x^2 + y^2)

75. The Taylor series expansion of f(x, y) = e^x * cos y about (0, 0) up to second degree terms is:

(A) 1 + x + (1/2)*(x^2 - y^2)
(B) 1 + x + y + x^2 + y^2
(C) 1 + x - y^2
(D) 1 + x + (1/2)*(x^2 + y^2)

76. If u = sin^(-1)((x^2 + y^2) / (x + y)), evaluate x^2*(d^2u/dx^2) + 2xy*(d^2u/dxdy) + y^2*(d^2u/dy^2):

(A) tan u * (tan^2 u - 1)
(B) tan^3 u
(C) tan u * sec^2 u
(D) tan u * (1 - tan^2 u)

77. If u = tan^(-1)((x^3 + y^3) / (x - y)), find x*(du/dx) + y*(du/dy):

(A) sin(2u)
(B) 2 * sin(2u)
(C) cos(2u)
(D) tan(2u)

78. If u = tan^(-1)((x^3 + y^3) / (x - y)), evaluate x^2*(d^2u/dx^2) + 2xy*(d^2u/dxdy) + y^2*(d^2u/dy^2):

(A) 2 * cos(3u) * sin u
(B) sin(4u) - sin(2u)
(C) 2 * sin(u) * cos(3u)
(D) sin(2u) * (2*cos(2u) - 1)

79. If u = f(r) where r^2 = x^2 + y^2 + z^2, prove that d^2u/dx^2 + d^2u/dy^2 + d^2u/dz^2 equals:

(A) f''(r) + (2/r)*f'(r)
(B) f''(r) + (1/r)*f'(r)
(C) f''(r) - (2/r)*f'(r)
(D) 3 * f''(r)

80. Find the maximum and minimum values of f(x, y) = x^2 + y^2 on the ellipse x^2/4 + y^2/9 = 1:

(A) Max = 9, Min = 4
(B) Max = 13, Min = 5
(C) Max = 36, Min = 1
(D) Max = 81, Min = 16

81. Find the dimensions of the rectangular box of maximum volume V = x*y*z that can be inscribed in the ellipsoid x^2/a^2 + y^2/b^2 + z^2/c^2 = 1:

(A) 2a/sqrt(3), 2b/sqrt(3), 2c/sqrt(3)
(B) a/3, b/3, c/3
(C) a/sqrt(2), b/sqrt(2), c/sqrt(2)
(D) a, b, c

82. The maximum volume of the rectangular box inscribed in the ellipsoid x^2/a^2 + y^2/b^2 + z^2/c^2 = 1 is:

(A) 8abc / (3 * sqrt(3))
(B) abc / (3 * sqrt(3))
(C) 4abc / 3
(D) abc

83. If x + y + z = u, y + z = u*v, z = u*v*w, find the Jacobian d(x, y, z)/d(u, v, w):

(A) u^2 * v
(B) u * v^2
(C) u^2 * v^2 * w
(D) u * v * w

84. If u = x*y/z, v = y*z/x, w = z*x/y, find the Jacobian d(u, v, w)/d(x, y, z):

(A) 1
(B) 2
(C) 4
(D) 8

85. Find the shortest distance from the point (1, 0, -2) to the plane x + 2y + z = 4:

(A) 5 / sqrt(6)
(B) sqrt(6)
(C) 6 / sqrt(5)
(D) 3

86. Find the point on the surface z = x*y + 1 closest to the origin (0, 0, 0):

(A) (0, 0, 1)
(B) (1, 1, 2)
(C) (0, 0, 0)
(D) (1, -1, 0)

87. Expand f(x, y) = x^2 * y + 3y - 2 in powers of (x - 1) and (y + 2) using Taylor's theorem. The constant term is:

(A) -6
(B) -8
(C) -4
(D) -2

88. Divide a positive number 'a' into three parts x, y, z such that their product x*y*z is a maximum. The parts are:

(A) a/3, a/3, a/3
(B) a/2, a/4, a/4
(C) a/6, a/3, a/2
(D) a/4, a/4, a/2

89. Find the maximum value of f(x, y) = sin x * sin y * sin(x + y) for 0 < x, y < pi:

(A) 3 * sqrt(3) / 8 (at x = y = pi/3)
(B) 1 / 8
(C) sqrt(3) / 4
(D) 1

90. In microbial batch fermentation, oxygen uptake rate OUR = kLa * (C_star - C_L). If kLa = 0.04 * (P/V)^0.7 * (v_s)^0.2, what is the percentage change in kLa for a 10% increase in power input P/V?

(A) 7%
(B) 10%
(C) 4%
(D) 14%

91. The error in calculating the area of a triangle from two sides a, b and included angle C when C has a small error delta-C (in radians) is:

(A) (1/2) * a * b * cos C * delta-C
(B) (1/2) * a * b * sin C * delta-C
(C) a * b * cos C
(D) (1/2) * delta-C

92. If u = x^2 - y^2 and v = 2xy, find (dx/du)_v:

(A) x / [2*(x^2 + y^2)]
(B) y / [2*(x^2 + y^2)]
(C) -x / (x^2 + y^2)
(D) 1 / (2x)

93. If z = f(x, y) satisfies x^2 + y^2 + z^2 = 1, then dz/dx is:

(A) -x / z
(B) x / z
(C) -z / x
(D) -y / z

94. If z satisfies x^2 + y^2 + z^2 = 1, then d^2z/dx^2 is:

(A) - (x^2 + z^2) / z^3 = - (1 - y^2) / z^3
(B) -1 / z
(C) -(z^2 - x^2) / z^3
(D) -x^2 / z^3

95. Find all stationary points of f(x, y) = 4x*y - x^4 - y^4:

(A) (0, 0), (1, 1), (-1, -1)
(B) (0, 0) only
(C) (1, -1), (-1, 1)
(D) (2, 2), (-2, -2)

96. For f(x, y) = 4x*y - x^4 - y^4, the points (1, 1) and (-1, -1) are both:

(A) Local Maxima with value 2
(B) Local Minima with value -2
(C) Saddle Points
(D) Inconclusive

97. For f(x, y) = 4x*y - x^4 - y^4, the origin (0, 0) is a:

(A) Saddle Point (r*t - s^2 = -16 < 0)
(B) Local Minimum
(C) Local Maximum
(D) Inflection point

98. A biotechnology company produces two recombinant enzymes with quantities x and y. Profit function is Pi(x, y) = 80x + 100y - x^2 - y^2 - xy. Maximum profit occurs at:

(A) x = 20, y = 40
(B) x = 30, y = 30
(C) x = 40, y = 20
(D) x = 10, y = 50

99. The maximum profit for Pi(x, y) = 80x + 100y - x^2 - y^2 - xy is:

(A) 2800
(B) 2600
(C) 3000
(D) 2400

100. If u = x + y + z, u*v = y + z, u*v*w = z, then x, y, z in terms of u, v, w are:

(A) x = u(1 - v), y = uv(1 - w), z = uvw
(B) x = u, y = v, z = w
(C) x = 1 - u, y = 1 - v, z = 1 - w
(D) x = uvw, y = uv, z = u

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Define the limit and continuity of a function of two variables f(x, y). Prove that the limit of f(x, y) = (x * y) / (x^2 + y^2) as (x, y) -> (0, 0) does not exist using path approaches.

Q2. State and prove Clairaut's (Schwarz's) Theorem on the equality of mixed second-order partial derivatives: f_xy = f_yx.

Q3. State and prove Euler's Theorem on Homogeneous Functions of degree n for two independent variables: x*(du/dx) + y*(du/dy) = n*u.

Q4. Prove the second-order extension of Euler's Theorem: If u(x, y) is homogeneous of degree n, then x^2*(d^2u/dx^2) + 2xy*(d^2u/dxdy) + y^2*(d^2u/dy^2) = n*(n - 1)*u.

Q5. If u = arcsin((x + y) / (sqrt(x) + sqrt(y))), prove that: (i) x*(du/dx) + y*(du/dy) = (1/2)*tan(u), and (ii) x^2*(d^2u/dx^2) + 2xy*(d^2u/dxdy) + y^2*(d^2u/dy^2) = -(1/4)*tan(u)*cos(2u) / cos^2(u).

Q6. If u = arctan((x^3 + y^3) / (x - y)), prove that: (i) x*(du/dx) + y*(du/dy) = sin(2u), and (ii) x^2*(d^2u/dx^2) + 2xy*(d^2u/dxdy) + y^2*(d^2u/dy^2) = 2*cos(3u)*sin(u).

Q7. Define the Total Differential of a function z = f(x, y) and derive the Chain Rule for composite functions z = f(x, y), where x = x(t) and y = y(t).

Q8. If z = f(u, v) where u = x^2 - y^2 and v = 2xy, prove that: (dz/dx)^2 + (dz/dy)^2 = 4 * (x^2 + y^2) * [(dz/du)^2 + (dz/dv)^2].

Q9. Derive the formulas for dy/dx and d^2y/dx^2 for an implicit function F(x, y) = 0. Apply these to compute dy/dx and d^2y/dx^2 for the Folium of Descartes x^3 + y^3 - 3axy = 0.

Q10. Define the Jacobian determinant for two and three variables. Give its geometric interpretation as a local magnification factor for areas and volumes.

Q11. State and prove the Chain Rule for Jacobians: If u, v are functions of x, y and x, y are functions of r, theta, then d(u, v)/d(r, theta) = [d(u, v)/d(x, y)] * [d(x, y)/d(r, theta)].

Q12. Prove the Reciprocal Property of Jacobians: J * J' = 1, where J = d(u, v)/d(x, y) and J' = d(x, y)/d(u, v). Verify this property for polar coordinates x = r*cos theta, y = r*sin theta.

Q13. State the criterion of functional dependence of two functions u(x, y) and v(x, y). Prove that a necessary and sufficient condition for u and v to be functionally dependent is that their Jacobian identically vanishes: d(u, v)/d(x, y) = 0.

Q14. Verify that the functions u = (x - y) / (x + y) and v = (x * y) / (x + y)^2 are functionally dependent. Find the explicit functional relation connecting u and v.

Q15. If x = u*(1 - v), y = u*v*(1 - w), and z = u*v*w, evaluate the Jacobian determinant: J = d(x, y, z) / d(u, v, w).

Q16. Derive Taylor's Theorem for a function of two variables f(x, y) about point (a, b). Express the expansion in operator form.

Q17. Expand f(x, y) = e^x * cos(y) in Maclaurin series about (0, 0) up to terms of degree 3.

Q18. Expand f(x, y) = x^2 * y + 3y - 2 in powers of (x - 1) and (y + 2) using Taylor's theorem.

Q19. State the necessary and sufficient conditions for local maxima, minima, and saddle points of a function of two variables f(x, y). Prove the discriminant criterion using the Hessian matrix.

Q20. Find and classify all stationary points of the function: f(x, y) = x^3 + y^3 - 3axy (where a > 0).

Q21. Find and classify all critical points of the function: f(x, y) = 4xy - x^4 - y^4.

Q22. Explain the theoretical foundation of Lagrange's Method of Undetermined Multipliers. Provide its geometric interpretation using level curves and gradient vectors.

Q23. Find the maximum and minimum values of f(x, y, z) = x^2 + y^2 + z^2 subject to the constraint x + y + z = 3a (a > 0) using Lagrange's method.

Q24. Find the dimensions of the rectangular box of maximum volume V = 8*x*y*z that can be inscribed inside the ellipsoid x^2/a^2 + y^2/b^2 + z^2/c^2 = 1.

Q25. Derive formulas for error estimation using total differentials. If the equivalent resistance R of two resistors in parallel is given by 1/R = 1/R1 + 1/R2, and R1, R2 have possible errors of 1% and 2% respectively, calculate the maximum percentage error in R when R1 = 100 Ohms and R2 = 200 Ohms.

Q26. Define the directional derivative of a scalar field f(x, y, z) along a direction u. Prove that the directional derivative is maximum in the direction of grad(f), and compute the directional derivative of f(x, y, z) = x^2 * y * z + 4*x*z^2 at point (1, -2, -1) in the direction of vector

Q27. Derive the equations of the tangent plane and normal line to a level surface F(x, y, z) = 0 at point P(x0, y0, z0). Find both equations for the ellipsoid 2*x^2 + 3*y^2 + z^2 = 21 at point (1, 2, 3).

Q28. In biotechnology, optimize the dilution rate D and inlet substrate concentration S0 to maximize steady-state biomass productivity P(D, S0) = D * X in a chemostat, where biomass X = Y * [S0 - (Ks * D) / (mu_max - D)]. Formulate the optimization problem using multivariate calculus.

Q29. In bio-thermodynamics, derive the four Maxwell Relations using Clairaut's theorem on the equality of mixed second-order partial derivatives of thermodynamic potentials (U, H, A, G).

Q30. A rectangular bioreactor cooling jacket has surface area S = 2*(xy + yz + zx) and fixed volume V = x*y*z = 1000 m^3. Use Lagrange's Method of Multipliers to find dimensions (x, y, z) that minimize heat loss surface area.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1:	[B]	**Q21:**	[C]	**Q41:**	[C]	**Q61:**	[B]	**Q81:**	[A]
Q2:	[B]	**Q22:**	[B]	**Q42:**	[B]	**Q62:**	[B]	**Q82:**	[A]
Q3:	[B]	**Q23:**	[A]	**Q43:**	[A]	**Q63:**	[A]	**Q83:**	[A]
Q4:	[B]	**Q24:**	[B]	**Q44:**	[A]	**Q64:**	[A]	**Q84:**	[C]
Q5:	[C]	**Q25:**	[B]	**Q45:**	[A]	**Q65:**	[A]	**Q85:**	[A]
Q6:	[B]	**Q26:**	[B]	**Q46:**	[B]	**Q66:**	[C]	**Q86:**	[A]
Q7:	[A]	**Q27:**	[B]	**Q47:**	[B]	**Q67:**	[A]	**Q87:**	[A]
Q8:	[B]	**Q28:**	[B]	**Q48:**	[B]	**Q68:**	[A]	**Q88:**	[A]
Q9:	[B]	**Q29:**	[B]	**Q49:**	[B]	**Q69:**	[A]	**Q89:**	[A]
Q10:	[B]	**Q30:**	[B]	**Q50:**	[A]	**Q70:**	[A]	**Q90:**	[A]
Q11:	[B]	**Q31:**	[A]	**Q51:**	[A]	**Q71:**	[A]	**Q91:**	[A]
Q12:	[B]	**Q32:**	[B]	**Q52:**	[C]	**Q72:**	[B]	**Q92:**	[A]
Q13:	[B]	**Q33:**	[B]	**Q53:**	[B]	**Q73:**	[A]	**Q93:**	[A]
Q14:	[A]	**Q34:**	[B]	**Q54:**	[A]	**Q74:**	[A]	**Q94:**	[A]
Q15:	[A]	**Q35:**	[B]	**Q55:**	[A]	**Q75:**	[A]	**Q95:**	[A]
Q16:	[B]	**Q36:**	[A]	**Q56:**	[A]	**Q76:**	[A]	**Q96:**	[A]
Q17:	[B]	**Q37:**	[A]	**Q57:**	[C]	**Q77:**	[A]	**Q97:**	[A]
Q18:	[B]	**Q38:**	[A]	**Q58:**	[B]	**Q78:**	[D]	**Q98:**	[A]
Q19:	[C]	**Q39:**	[B]	**Q59:**	[B]	**Q79:**	[A]	**Q99:**	[A]
Q20:	[C]	**Q40:**	[B]	**Q60:**	[A]	**Q80:**	[A]	**Q100:**	[A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

- By the Mean Value Theorem, there exists c1 in (a, a + h) such that:
Delta(h, k) = phi'(c1) * h = h * [fx(c1, b + k) - fx(c1, b)].
- Now define g(y) = fx(c1, y). Applying the Mean Value Theorem to g on [b, b + k]:
fx(c1, b + k) - fx(c1, b) = g'(d1) * k = f_yx(c1, d1) * k, for some d1 in (b, b + k).
- Substituting back:
Delta(h, k) = h * k * f_yx(c1, d1). (Eq. 1)
- Now reverse the order of grouping: define psi(y) = f(a + h, y) - f(a, y).
Then: Delta(h, k) = psi(b + k) - psi(b).
- By LMVT on psi, there exists d2 in (b, b + k) such that:
Delta(h, k) = psi'(d2) * k = k * [fy(a + h, d2) - fy(a, d2)].
- Applying LMVT to h(x) = fy(x, d2) on [a, a + h] gives c2 in (a, a + h) such that:
fy(a + h, d2) - fy(a, d2) = f_xy(c2, d2) * h.
- Substituting back:
Delta(h, k) = h * k * f_xy(c2, d2). (Eq. 2)
- Equating Eq. 1 and Eq. 2 and dividing by h * k != 0:
f_yx(c1, d1) = f_xy(c2, d2).
- Take limits as (h, k) -> (0, 0).
Since c1, c2 -> a and d1, d2 -> b, and since both fxy and fyx are continuous at (a, b):
f_yx(a, b) = f_xy(a, b).
Hence proved.

>>Solution to Question 3: State and prove Euler's Theorem on Homogeneous Functions of degree n for two independent variables: x*(du/dx) + y*(du/dy) = n*u.

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x^2*(d^2u/dx^2) + 2xy*(d^2u/dxdy) + y^2*(d^2u/dy^2) = G(u),
where G(u) = F(u) * [F'(u) - 1].
- Compute F'(u):
F'(u) = d/du [(1/2)*tan(u)] = (1/2) * sec^2(u).
- Compute G(u):
G(u) = (1/2)*tan(u) * [(1/2)*sec^2(u) - 1]
= (1/2)*tan(u) * [(1 - 2*cos^2(u)) / (2*cos^2(u))]
= (1/4)*tan(u) * [-cos(2u) / cos^2(u)]
= - (1/4) * tan(u) * cos(2u) / cos^2(u). (Part ii proved).

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where epsilon1, epsilon2 -> 0 as (Delta_x, Delta_y) -> (0, 0).
- The linear principal part is called the Total Differential dz:
dz = (df/dx) * dx + (df/dy) * dy.
2. Derivation of Chain Rule for Total Derivative dz/dt:
- Suppose z = f(x, y) is differentiable and x = x(t), y = y(t) are differentiable functions of t.
- When t changes by Delta_t, x and y change by:
Delta_x = x(t + Delta_t) - x(t), Delta_y = y(t + Delta_t) - y(t).
- Corresponding increment in z:
Delta_z = (df/dx) * Delta_x + (df/dy) * Delta_y + epsilon1 * Delta_x + epsilon2 * Delta_y.
- Divide throughout by Delta_t != 0:
Delta_z / Delta_t = (df/dx) * (Delta_x / Delta_t) + (df/dy) * (Delta_y / Delta_t) + epsilon1 * (Delta_x / Delta_t) + epsilon2 * (Delta_y / Delta_t).
- Take the limit as Delta_t -> 0:
Since x(t) and y(t) are differentiable, they are continuous, so Delta_x -> 0 and Delta_y -> 0 as Delta_t -> 0.
Consequently, epsilon1 -> 0 and epsilon2 -> 0.
Also: lim (Delta_x / Delta_t) = dx / dt, and lim (Delta_y / Delta_t) = dy / dt.
- Therefore:
dz / dt = (df/dx) * (dx/dt) + (df/dy) * (dy/dt). (Proved).

>>Solution Step Question 8: (the formulas for dx/dx^2 and d^2y/dx^2 to prove that d/dx(dz/dx) = d^2z/dx^2 + (dz/dy)^2 * d^2y/dx^2 + 2 * (dz/dy) * (dz/dx) * d^2y/dx^2 for the Foli

- Let F(x, y) = 0 define y implicitly as a differentiable function of x.
- Taking the total differential: dF = Fx * dx + Fy * dy = 0.
- Solving for dy/dx:
dy / dx = - Fx / Fy (provided Fy != 0).
- Second derivative d^2y / dx^2:
d^2y / dx^2 = d/dx [- Fx / Fy] = - [Fy * d/dx(Fx) - Fx * d/dx(Fy)] / (Fy)^2.
By chain rule:
d/dx(Fx) = Fxx + Fxy * (dy/dx) = Fxx - Fxy * (Fx / Fy) = (Fxx * Fy - Fxy * Fx) / Fy.
d/dx(Fy) = Fyx + Fyy * (dy/dx) = Fyx - Fyy * (Fx / Fy) = (Fxy * Fy - Fyy * Fx) / Fy.
Substitute into numerator:
d^2y / dx^2 = - [Fy * (Fxx*Fy - Fxy*Fx) - Fx * (Fxy*Fy - Fyy*Fx)] / (Fy)^3
= - [Fxx * Fy^2 - 2 * Fxy * Fx * Fy + Fyy * Fx^2] / (Fy)^3. (Proved).
2. Application to Folium of Descartes: F(x, y) = x^3 + y^3 - 3axy = 0:
- Compute partial derivatives:
Fx = 3x^2 - 3ay = 3*(x^2 - ay).
Fy = 3y^2 - 3ax = 3*(y^2 - ax).
Fxx = 6x, Fxy = -3a, Fyy = 6y.
- First derivative dy/dx:
dy / dx = - Fx / Fy = - [3*(x^2 - ay)] / [3*(y^2 - ax)]
= (ay - x^2) / (y^2 - ax).
- Second derivative d^2y/dx^2:
Numerator N = 6x*(3(y^2 - ax))^2 - 2*(-3a)*(3(x^2 - ay))*(3(y^2 - ax)) + 6y*(3(x^2 - ay))^2
= 54 * [x*(y^2 - ax)^2 + a*(x^2 - ay)*(y^2 - ax) + y*(x^2 - ay)^2].
Expanding and using x^3 + y^3 = 3axy simplifies the bracket to: -2a^3*xy.
Therefore:
d^2y / dx^2 = - 2a^3 * x * y / (y^2 - ax)^3.

>> **Solution to Question 10:** Define the Jacobian determinant for two and three variables. Give its geometric interpretation as a local magnification factor for areas a

= (cos theta)*(r * cos theta) - (-r * sin theta)*(sin theta)
= r * cos^2 theta + r * sin^2 theta = r * (cos^2 theta + sin^2 theta) = r.
- Inverse Transformation: r = sqrt(x^2 + y^2), theta = arctan(y / x).
- Compute J' = d(r, theta) / d(x, y):
dr/dx = x / sqrt(x^2 + y^2) = (r * cos theta) / r = cos theta.
dr/dy = y / sqrt(x^2 + y^2) = (r * sin theta) / r = sin theta.
dtheta/dx = -y / (x^2 + y^2) = (-r * sin theta) / r^2 = -sin theta / r.
dtheta/dy = x / (x^2 + y^2) = (r * cos theta) / r^2 = cos theta / r.
J' = det([cos theta, sin theta; -sin theta / r, cos theta / r])
= (cos theta)*(cos theta / r) - (sin theta)*(-sin theta / r)
= (cos^2 theta + sin^2 theta) / r = 1 / r.
- Verify product:
J * J' = r * (1 / r) = 1.
Confirmed.

Solution to Question 12: the criterion of functional dependence of two functions u(x, y) and v(x, y). Prove that a necessary and sufficient condition for u and

Hence proved.

>>Solution Statement: Let the functions u = wx and z = w*v and evaluate the Jacobian determinant. Find the explicit functional relation con

dz/dw = uv.
Step 4: Set up the Jacobian Matrix:
[1 - v -u 0]
J = [v*(1 - w) u*(1 - w) -uv]
[vw uw uv]
Step 5: Apply column operations to simplify:
Add Row 3 to Row 2:
Row 2 + Row 3:
[v*(1 - w) + vw, u*(1 - w) + uw, -uv + uv] = [v, u, 0].
The matrix becomes:
[1 - v -u 0]
J = [v u 0]
[vw uw uv]
Step 6: Expand the determinant along column 3:
det(J) = uv * det([1 - v, -u; v, u]).
Step 7: Evaluate the 2x2 determinant:
det([1 - v, -u; v, u]) = (1 - v)*u - (-u)*v = u - uv + uv = u.
Step 8: Final multiplication:
det(J) = uv * u = u^2 * v.
Conclusion:
d(x, y, z) / d(u, v, w) = u^2 * v.

Solution to Question 15: State Taylor's Theorem for a function of two variables f(x, y) about point (a, b). Express the expansion in operator form.

>>Solution Question 15 Find f(x, y) = e^2 * cos(3y) in Maclaurin's series about (0, 0) upto terms of third degree.

Step 3: Construct Taylor Polynomial:
f(x, y) = f(1, -2) + [h*fx + k*fy] + (1/2!)*[h^2*fxx + 2hk*fxy + k^2*fyy]
+ (1/3!)*[3h^2*k*fxxy]
Substitute values:
f(x, y) = -10 + [-4*(x - 1) + 4*(y + 2)]
+ (1/2)*[-4*(x - 1)^2 + 4*(x - 1)*(y + 2)]
+ (1/6)*[3*(2)*(x - 1)^2 * (y + 2)]
= -10 - 4*(x - 1) + 4*(y + 2) - 2*(x - 1)^2 + 2*(x - 1)*(y + 2) + (x - 1)^2 * (y + 2).
Conclusion:
f(x, y) = -10 - 4(x - 1) + 4(y + 2) - 2(x - 1)^2 + 2(x - 1)(y + 2) + (x - 1)^2(y + 2).

>>Solution Step Question 20: Derive the classical second derivative test for local maxima, minima, and saddle points of a function of two variables f(x, y). Prove th

fx = 3x^2 - 3ay = 3*(x^2 - ay)
fy = 3y^2 - 3ax = 3*(y^2 - ax).
Step 2: Stationary Points (fx = 0 and fy = 0):
From fx = 0: x^2 = ay => y = x^2 / a.
Substitute into fy = 0:
3 * [(x^2 / a)^2 - ax] = 0
=> x^4 / a^2 - ax = 0 => x^4 - a^3 * x = 0
=> x * (x^3 - a^3) = 0.
Real roots: x = 0 and x = a.
- If x = 0: y = 0^2 / a = 0. Stationary point P1(0, 0).
- If x = a: y = a^2 / a = a. Stationary point P2(a, a).
Step 3: Second Partial Derivatives:
r = fxx = 6x
s = fxy = -3a
t = fyy = 6y
Discriminant: Delta = r * t - s^2 = (6x)*(6y) - (-3a)^2 = 36xy - 9a^2.
Step 4: Classification of Point P1(0, 0):
At (0, 0):
r = 0, s = -3a, t = 0.
Delta = 36(0)(0) - 9a^2 = -9a^2.
Since a > 0, Delta = -9a^2 < 0.
Therefore, the origin P1(0, 0) is a SADDLE POINT.
Step 5: Classification of Point P2(a, a):
At (a, a):
r = 6a > 0 (since a > 0).
s = -3a.
t = 6a.
Delta = (6a)*(6a) - (-3a)^2 = 36a^2 - 9a^2 = 27a^2 > 0.
Since Delta = 27a^2 > 0 and r = 6a > 0:
f attains a strict LOCAL MINIMUM at P2(a, a).
Minimum value:
f_min = f(a, a) = a^3 + a^3 - 3a(a)(a) = 2a^3 - 3a^3 = -a^3.

Solution to Question 21

>> Solution to Question 21: Find and classify all critical points of the function: f(x, y) = 4xy - x^4 - y^4.

s = fxy = 4
t = fyy = -12y^2
Discriminant: Delta = r * t - s^2 = (-12x^2)*(-12y^2) - 16 = 144 * x^2 * y^2 - 16.
Step 4: Classification of Point P1(0, 0):
r = 0, t = 0, s = 4.
Delta = 144(0) - 16 = -16 < 0.
P1(0, 0) is a SADDLE POINT. f(0, 0) = 0.
Step 5: Classification of Point P2(1, 1):
r = -12(1)^2 = -12 < 0.
t = -12(1)^2 = -12.
s = 4.
Delta = 144(1)(1) - 16 = 128 > 0.
Since Delta > 0 and r < 0:
P2(1, 1) is a LOCAL MAXIMUM.
f_max = 4(1)(1) - 1^4 - 1^4 = 4 - 2 = 2.
Step 6: Classification of Point P3(-1, -1):
r = -12(-1)^2 = -12 < 0.
t = -12.
s = 4.
Delta = 144(1)(1) - 16 = 128 > 0.
Since Delta > 0 and r < 0:
P3(-1, -1) is also a LOCAL MAXIMUM.
f_max = 4(-1)(-1) - (-1)^4 - (-1)^4 = 4 - 2 = 2.

>> Solution to Question 22

Explain the theoretical foundation of Lagrange's Method of Undetermined Multipliers. Provide its geometric interpretation using level cu

- Solving this system of 4 equations in 4 unknowns (x, y, z, lambda) yields the critical points.

-2*lambda * (x^2 / a^2) = -2*lambda * (y^2 / b^2) = -2*lambda * (z^2 / c^2).
Since V > 0, lambda != 0:
x^2 / a^2 = y^2 / b^2 = z^2 / c^2.
Step 5: Substitute into Ellipsoid Constraint:
(x^2 / a^2) + (x^2 / a^2) + (x^2 / a^2) = 1
=> 3 * (x^2 / a^2) = 1 => x^2 / a^2 = 1/3
=> x = a / sqrt(3).
Similarly:
y = b / sqrt(3), z = c / sqrt(3).
Step 6: Dimensions and Maximum Volume:
- Dimensions of the box:
Length = 2x = 2a / sqrt(3).
Width = 2y = 2b / sqrt(3).
Height = 2z = 2c / sqrt(3).
- Maximum Volume:
V_max = 8 * [a / sqrt(3)] * [b / sqrt(3)] * [c / sqrt(3)]
= 8*a*b*c / (3 * sqrt(3)).
Conclusion:
The dimensions are 2a/sqrt(3), 2b/sqrt(3), 2c/sqrt(3), and maximum volume is 8abc / (3*sqrt(3)).

>>Solution to Question 25: Derive formulas for error estimation using total differentials. If the equivalent resistance R of two resistors in parallel is given by 1/R = 1/

>>Solution Statement Question 25: Find the directional derivative of a scalar field in a normal direction. Prove that the direction derivative is maximum in the

4 * (x - 1) + 12 * (y - 2) + 6 * (z - 3) = 0
4x - 4 + 12y - 24 + 6z - 18 = 0
4x + 12y + 6z = 46.
Divide by 2:
2x + 6y + 3z = 23.
- Normal Line:
(x - 1) / 4 = (y - 2) / 12 = (z - 3) / 6,
or dividing direction ratios by 2:
(x - 1) / 2 = (y - 2) / 6 = (z - 3) / 3.

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1. Thermodynamic Potentials and Fundamental Equations:
- Internal Energy U(S, V): dU = T*dS - P*dV.
- Enthalpy H(S, P): dH = T*dS + V*dP.
- Helmholtz Free Energy A(T, V): dA = -S*dT - P*dV.
- Gibbs Free Energy G(T, P): dG = -S*dT + V*dP.
2. Derivation of Maxwell Relation 1 (from U):
- From dU = T*dS - P*dV:
(dU/dS)_V = T and (dU/dV)_S = -P.
- By Clairaut's Theorem, mixed partials commute: d^2U / dVdS = d^2U / dSdV.
d/dV [(dU/dS)_V]_S = d/dS [(dU/dV)_S]_V
=> (dT / dV)_S = - (dP / dS)_V. (Relation 1).
3. Derivation of Maxwell Relation 2 (from H):
- From dH = T*dS + V*dP:
(dH/dS)_P = T and (dH/dP)_S = V.
- Equating mixed partials: d^2H / dPdS = d^2H / dSdP:
d/dP [(dH/dS)_P]_S = d/dS [(dH/dP)_S]_P
=> (dT / dP)_S = (dV / dS)_P. (Relation 2).
4. Derivation of Maxwell Relation 3 (from A):
- From dA = -S*dT - P*dV:
(dA/dT)_V = -S and (dA/dV)_T = -P.
- Equating mixed partials: d^2A / dVdT = d^2A / dTdV:
d/dV [-S]_T = d/dT [-P]_V
=> (dS / dV)_T = (dP / dT)_V. (Relation 3).
5. Derivation of Maxwell Relation 4 (from G):
- From dG = -S*dT + V*dP:
(dG/dT)_P = -S and (dG/dP)_T = V.
- Equating mixed partials: d^2G / dPdT = d^2G / dTdP:
d/dP [-S]_T = d/dT [V]_P
=> - (dS / dP)_T = (dV / dT)_P, or (dS / dP)_T = - (dV / dT)_P. (Relation 4).
Conclusion:
Maxwell's relations permit the calculation of experimentally inaccessible entropy changes (dS/dP) from measurable volumetric thermal expansion coefficients (dV/dT).

>>Solution to Question 30: A rectangular bioreactor cooling jacket has surface area S = 2*(xy + yz + zx) and fixed volume V = x*y*z = 1000 m^3. Use Lagrange's M

2*(1/z + 1/y) = 2*(1/z + 1/x) => 1/y = 1/x => x = y.
2*(1/z + 1/x) = 2*(1/y + 1/x) => 1/z = 1/y => y = z.
Therefore, by symmetry:
x = y = z (a perfect cube!).
Step 6: Solve for dimensions using the constraint:
x * y * z = 1000
=> x^3 = 1000 => x = (1000)^(1/3) = 10 m.
Thus:
x = 10 m, y = 10 m, z = 10 m.
Step 7: Minimum Surface Area:
S_min = 2 * [10(10) + 10(10) + 10(10)] = 2 * [100 + 100 + 100] = 2 * 300 = 600 m^2.
Conclusion:
The heat loss surface area is minimized when the bioreactor jacket is a cube with dimensions 10 m x 10 m x 10 m, yielding a minimum surface area of 600 m^2.
